



# Crash Course in Maths for Cognitive Science

## Session 2 | Statistics & Inference

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## What today is about

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Last session we studied *single* uncertain quantities. Today we turn to *data*: many observations at once, and what we can learn from them.

- Summarise data with a few numbers: *mean, variance, covariance*.
- Understand why averaging works: the *Law of Large Numbers* and the *Central Limit Theorem*.
- Infer from a sample to a population: *standard errors, confidence intervals, hypothesis tests*.
- Build our first real model: *linear regression*.

As in Session 1, we stop to *solve examples together* at every step.

# The thread running through everything

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## Population vs. sample

There is a fixed but unknown truth about the whole *population* (for example, the true average reaction time of all humans). We only ever see a limited *sample*. Statistics is the disciplined art of guessing the population from the sample, and honestly reporting how uncertain that guess is.

Keep asking, on every slide: is this a statement about the population, or about our particular sample?

## The centre of a distribution: expectation

The first thing we want to know about data is where it is centred.

### Definition

The **expectation** (mean) is the probability-weighted average value:

$$\mathbb{E}[X] = \sum_x x p(x) \quad (\text{discrete}), \quad \mathbb{E}[X] = \int x f(x) dx \quad (\text{continuous}).$$

Picture the distribution as a pile of mass on the number line: the expectation is its *balance point*. It need not be a value the variable can actually take, a fair die has mean 3.5.

## Expectation adds up nicely

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The property that makes expectation so useful is *linearity*, and it holds always, even when variables influence each other.

### Useful fact

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c.$$

### Intuition

This is why we can rescale, shift, and combine measures freely and still track their average, no need to re-derive the whole distribution.

## Worked example 1: rescaling a score

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### Example

A fair die  $X$  is rolled, and a game pays  $Y = 2X + 1$  points.

1. What is  $\mathbb{E}[X]$ ?
2. What is the expected payout  $\mathbb{E}[Y]$ ?

*Try it yourself first — the solution is on the next slide.*

## Worked example 1: solution

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1.  $\mathbb{E}[X] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = 3.5$ .
2. By linearity,  $\mathbb{E}[Y] = 2\mathbb{E}[X] + 1 = 2(3.5) + 1 = 8$ , no need to average the six payouts separately.

Linearity turns a distribution-level question into one line of arithmetic. It will reappear constantly (e.g. mean-centring data subtracts  $\mathbb{E}[X]$  from every value).

## How spread out is the data? Variance

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### Definition

The **variance** is the average squared distance from the mean; the **standard deviation** is its square root (back in the original units):

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2], \quad \text{SD}(X) = \sqrt{\text{Var}(X)}.$$

We square the deviations so positive and negative ones do not cancel, and to penalise large departures more. The square root brings us back to sensible units, milliseconds, not milliseconds-squared.

## Three variance rules worth memorising

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### Useful fact

**Computational formula:**  $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ , usually fastest by hand.

**Shift and scale:**  $\text{Var}(aX + b) = a^2 \text{Var}(X)$ , adding a constant does not change spread; scaling multiplies the SD by  $|a|$ .

**Sum:**  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$ .

That last term, the covariance, is exactly the correction for when  $X$  and  $Y$  move together. It vanishes only when they are uncorrelated.

## Worked example 2: variance of a die

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### Example

Find the variance and standard deviation of a fair die, using the computational formula  $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ .

*Try it yourself first — the solution is on the next slide.*

## Worked example 2: solution

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The mean of the square:

$$\mathbb{E}[X^2] = \frac{1}{6}(1 + 4 + 9 + 16 + 25 + 36) = \frac{91}{6} \approx 15.17.$$

So

$$\text{Var}(X) = 15.17 - 3.5^2 = 15.17 - 12.25 = 2.92, \quad \text{SD}(X) = \sqrt{2.92} \approx 1.71.$$

A die roll typically lands about 1.7 away from its mean of 3.5.

## Worked example 3: spread of a neuron's firing

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### Example

A neuron's firing rate (spikes/s) is measured on five trials:

6, 8, 10, 12, 14.

Compute the mean, the variance (average squared deviation), and the standard deviation.

*Try it yourself first — the solution is on the next slide.*

## Worked example 3: solution

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Mean =  $\frac{6+8+10+12+14}{5} = 10$ . Deviations  $-4, -2, 0, 2, 4$ ; their squares  $16, 4, 0, 4, 16$  sum to 40. So

$$\text{Var} = \frac{40}{5} = 8, \quad \text{SD} = \sqrt{8} \approx 2.83.$$

### Intuition

The SD ( $\approx 2.8$  spikes/s) is in the same units as the data, so it is directly interpretable, unlike the variance ( $8 \text{ spikes}^2/\text{s}^2$ ). This is why error bars and “noise levels” are almost always quoted as SDs.

## How two variables move together

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### Definition

The **covariance** measures whether two quantities tend to be large or small together:

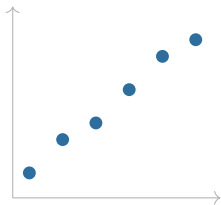
$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)],$$

and the **correlation** rescales it into  $[-1, 1]$ :  $\rho = \frac{\text{Cov}(X, Y)}{\text{SD}(X) \text{SD}(Y)}.$

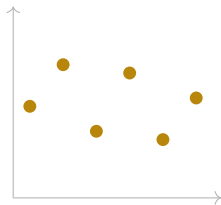
Covariance is positive when the two are usually on the same side of their means, negative when one is high while the other is low.

## What correlation looks like

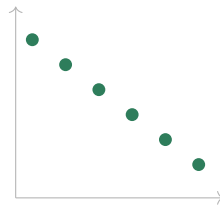
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$$\rho \approx +0.9$$



$$\rho \approx 0$$



$$\rho \approx -0.9$$

Correlation measures the strength and direction of a *linear* trend:  $+1$  a perfect rising line,  $0$  no linear trend,  $-1$  a perfect falling line.

## Useful facts about correlation

### Useful fact

- $\rho$  is *unit-free* and always lies in  $[-1, 1]$ .
- It is unchanged by shifting or positively rescaling either variable (mm or cm, °C or °C+5 give the same  $\rho$ ).
- $\rho = \pm 1$  exactly when the points lie on a perfect straight line.

### Key idea

Collecting all pairwise covariances of many variables gives the **covariance matrix**, the single object at the heart of PCA (Session 4). A statistical quantity becomes a geometric one.

## Worked example 4: do two neurons co-vary?

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### Example

Across five trials, neuron  $A$ 's rate is  $x = (1, 2, 3, 4, 5)$  and neuron  $B$ 's rate is  $y = (2, 4, 5, 4, 5)$  (arbitrary units); the means are  $\bar{x} = 3$ ,  $\bar{y} = 4$ .

Compute the correlation  $\rho$  between the two neurons. (*Helpful sums to find:  $\sum(x - \bar{x})(y - \bar{y})$ ,  $\sum(x - \bar{x})^2$ ,  $\sum(y - \bar{y})^2$ .*)

*Try it yourself first — the solution is on the next slide.*

## Worked example 4: solution

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The deviation products give

$$\sum (x - \bar{x})(y - \bar{y}) = 6, \quad \sum (x - \bar{x})^2 = 10, \quad \sum (y - \bar{y})^2 = 6.$$

So

$$\rho = \frac{6}{\sqrt{10 \times 6}} = \frac{6}{\sqrt{60}} \approx 0.77 :$$

a strong positive association, the two neurons tend to be high or low together (a *noise correlation*). We reuse these very sums for the regression line at the end of today.

## Worked example 5: guess the sign

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### Example

For each pair, is the correlation *positive*, *negative*, or *roughly zero*?

1. Hours of sleep vs. reaction time (more sleep  $\rightarrow$  faster responses).
2. Stimulus contrast vs. a neuron's firing rate.
3. A person's height vs. their exam score.

*Try it yourself first — the solution is on the next slide.*

## Worked example 5: solution

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1. **Negative:** more sleep  $\rightarrow$  *lower* reaction time.
2. **Positive:** higher contrast  $\rightarrow$  higher firing rate.
3. **Roughly zero:** no reason for a linear link.

### Common pitfall

$\rho = 0$  means no *linear* relationship, not “no relationship”, points on  $y = x^2$  have  $\rho = 0$  yet are perfectly predictable. And correlation is never, by itself, evidence of *causation*: a strong  $\rho$  is an invitation to investigate, not a conclusion about cause.

### Practice — try it

A variable  $X$  takes the values  $\{2, 4, 6\}$ , each with probability  $\frac{1}{3}$ .

1. Find  $\mathbb{E}[X]$ .
2. Find  $\text{Var}(X)$  using the computational formula.
3. If  $Y = 3X$ , what is  $\text{SD}(Y)$ ?

## Two theorems behind all of statistics

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Why can we trust an average of noisy measurements? Two results answer this.

### Key idea

**Law of Large Numbers:** as the sample size  $n$  grows, the sample mean  $\bar{X}_n$  converges to the true mean  $\mu$ . *Averages settle down.*

**Central Limit Theorem:** for large  $n$ , the sample mean is approximately Normal,  $\bar{X}_n \approx \mathcal{N}(\mu, \sigma^2/n)$ , *whatever the shape of the original data. Averages become bell-shaped.*

# The Central Limit Theorem, in pictures

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$n = 1$  (one die)



$n = 2$  (average of two)



large  $n$  (bell-shaped)

One die is *flat*; the average of two is already *triangular*; average enough and the shape is *Normal*, no matter what you started with.

## Why the Central Limit Theorem is remarkable

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The Law of Large Numbers is intuitive: more data, better estimate. The Central Limit Theorem says something stronger.

Even if individual observations are wildly non-Normal, a skewed reaction time, a 0/1 correct response, their *average* traces out a smooth Normal curve, with spread shrinking like  $\sigma/\sqrt{n}$ .

### Link to cognitive science

This underwrites *trial averaging* in EEG and fMRI: averaging  $n$  noisy trials cuts the noise SD by  $\sqrt{n}$ , pulling a buried event-related signal out of the noise. It also explains why so many measured traits look Gaussian, each is a sum of many small independent contributions.

## The $\sqrt{n}$ law of averaging

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### Useful fact

If  $X_1, \dots, X_n$  are independent with mean  $\mu$  and SD  $\sigma$ , the sample mean  $\bar{X}$  has

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

The key consequence is a *diminishing return*: to *halve* the uncertainty of an estimate you need *four times* as much data; to cut it tenfold, a hundredfold. This is why large studies are expensive.

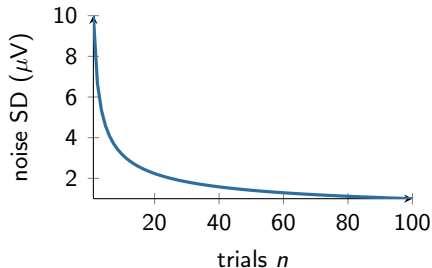
## Worked example 6: how many EEG trials?

### Example

Single-trial EEG responses carry noise with SD  $\sigma = 10 \mu\text{V}$ .

1. What is the noise SD after averaging 25 trials?
2. How many trials are needed to bring it down to  $1 \mu\text{V}$ ?

*Try it yourself first — the solution is on the next slide.*



## Worked example 6: solution

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1.  $SD = \frac{10}{\sqrt{25}} = \frac{10}{5} = 2 \mu V.$
2. Solve  $\frac{10}{\sqrt{n}} = 1 \Rightarrow \sqrt{n} = 10 \Rightarrow n = 100$ , four times as many trials as the 25 that gave  $2 \mu V.$

Every halving of the noise costs a *quadrupling* of trials, the practical face of the  $\sqrt{n}$  law, and why long recording sessions have sharply diminishing returns.

### Definition

A **parameter** is a fixed unknown describing the population (e.g. the true mean  $\mu$ ). An **estimator** is a recipe applied to the sample (e.g. the sample mean  $\bar{X}$ ). It is **unbiased** if it is right *on average* across many samples:  $\mathbb{E}[\hat{\theta}] = \theta$ .

## Why we divide by $n - 1$

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### Common pitfall

The sample variance divides by  $n - 1$ , not  $n$ :  $s^2 = \frac{1}{n - 1} \sum_{i=1}^n (x_i - \bar{x})^2$ .

Using the sample mean  $\bar{x}$  (instead of the unknown true mean) makes the deviations come out slightly too small; dividing by  $n - 1$  corrects exactly for this, giving an unbiased estimate. Know both the rule and the reason.

## Standard deviation vs. standard error

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This distinction is subtle and constantly misused, so it gets its own slide.

### Common pitfall

The **standard deviation**  $s$  describes the spread of the raw data; it does *not* shrink with more data. The **standard error**  $SE = s/\sqrt{n}$  describes the precision of our *estimate of the mean*; it *does* shrink as  $n$  grows.

When a figure shows error bars, always ask which they are: the spread of individuals, or the uncertainty in the mean? They tell very different stories.

## Worked example 7: which one shrinks?

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### Example

A population of reaction times has SD  $\sigma = 20$  ms.

1. Compute the standard error of the mean for a sample of  $n = 25$ .
2. And for  $n = 100$ .
3. As  $n$  grows from 25 to 100, what happened to the SD of the data, and to the SE of the mean?

*Try it yourself first — the solution is on the next slide.*

## Worked example 7: solution

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1.  $SE = \frac{20}{\sqrt{25}} = 4 \text{ ms.}$

2.  $SE = \frac{20}{\sqrt{100}} = 2 \text{ ms.}$

3. The *SD* stays 20 ms (it is a property of the data, not the sample size). The *SE* halved (from 4 to 2 ms): we know the mean twice as precisely with four times the data.

### Definition

A 95% confidence interval for the mean is  $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$ .

The width  $1.96 \sigma / \sqrt{n}$  comes straight from the Central Limit Theorem and the 95% mark of the Normal curve. The recipe is built so that, across many repeated samples, 95% of the intervals it produces contain the true  $\mu$ .

## When $\sigma$ is unknown: use $t$ , not $z$

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### Useful fact

In practice we estimate  $\sigma$  by the sample SD  $s$ . This extra uncertainty means the standardised mean follows *Student's  $t$ -distribution* with  $n - 1$  degrees of freedom, like the Normal, but with heavier tails.

For small samples the 95% multiplier exceeds 1.96 (about 2.26 at  $n = 10$ ), and it shrinks toward 1.96 as  $n$  grows. Rule of thumb: once  $n \gtrsim 30$ , the difference hardly matters.

## Worked example 8: a confidence interval for reaction time

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### Example

Over  $n = 64$  trials the mean reaction time is  $\bar{x} = 540$  ms; assume the population SD is  $\sigma = 80$  ms.

1. Give the 95% confidence interval for the true mean (use 1.96).
2. Give the 99% interval (use 2.576). Which is wider, and why?

*Try it yourself first — the solution is on the next slide.*

## Worked example 8: solution

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The standard error is  $SE = \frac{80}{\sqrt{64}} = \frac{80}{8} = 10$  ms.

- 95%:  $540 \pm 1.96(10) = 540 \pm 19.6 = [520.4, 559.6]$  ms.
- 99%:  $540 \pm 2.576(10) = 540 \pm 25.8 = [514.2, 565.8]$  ms.

The 99% interval is *wider*: demanding more confidence means casting a wider net.  
Confidence and precision trade off.

## What “95% confidence” really means

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### Common pitfall

A 95% confidence interval does *not* mean “there is a 95% probability the true mean lies in *this* interval”. In the classical framework  $\mu$  is fixed; it is the interval that is random.

The correct reading is about the *procedure*: repeat the experiment many times and 95% of the intervals you build this way would capture the truth.

### Practice — try it

1. A sample of  $n = 100$  has mean  $\bar{x} = 52$  with known  $\sigma = 10$ . Give a 95% confidence interval for  $\mu$ .
2. Testing  $H_0 : \mu = 50$  against  $\mu \neq 50$ , what is the z-statistic and the rough conclusion?
3. A treatment raises scores by 2 points, with within-group SD 10. What is Cohen's  $d$ , and is it small, medium, or large?

# The logic of a hypothesis test

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## Four steps

- (1) State a **null hypothesis**  $H_0$  (“nothing is going on”) and an alternative  $H_1$ .
- (2) Choose a **test statistic** that would be extreme if  $H_1$  were true.
- (3) Compute the  **$p$ -value**: assuming  $H_0$ , the probability of a result at least this extreme.
- (4) If  $p < \alpha$  (often 0.05), **reject**  $H_0$ .

## Worked example 9: is the coin fair?

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### Example

A coin lands heads 60 times in 100 flips. Test  $H_0 : p = 0.5$  against  $p \neq 0.5$ .

*Hint:* under  $H_0$  the number of heads has mean  $np = 50$  and standard deviation  $\sqrt{np(1-p)} = \sqrt{25} = 5$ .

*Try it yourself first — the solution is on the next slide.*

## Worked example 9: solution

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The standardised statistic:

$$z = \frac{60 - 50}{5} = 2.0.$$

A result at least this extreme (either tail) has probability  $\mathbb{P}(|Z| \geq 2.0) \approx 0.046$ . Since  $0.046 < 0.05$ , we **reject**  $H_0$ : weak but statistically significant evidence of bias.

## Worked example 10: is the subject above chance?

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### Link to cognitive science

Testing whether accuracy beats chance is the most common test in psychophysics.

### Example

A subject is correct on 44 of 60 trials in a yes/no task where chance is 0.5. Under  $H_0$  the number correct has mean 30 and SD  $\sqrt{60 \cdot 0.5 \cdot 0.5} = \sqrt{15} \approx 3.87$ . Compute the z-statistic and state the conclusion.

*Try it yourself first — the solution is on the next slide.*

## Worked example 10: solution

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$$z = \frac{44 - 30}{3.87} = \frac{14}{3.87} \approx 3.6.$$

This is far beyond 1.96 ( $p < 0.001$ ), so the subject performs *well above chance*. A large  $z$  here reflects both a real effect and a decent number of trials.

## Two ways to be wrong

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### Definition

A **Type I error** rejects a true  $H_0$  (a false positive), with probability  $\alpha$ . A **Type II error** fails to reject a false  $H_0$  (a false negative), with probability  $\beta$ . The **power** is  $1 - \beta$ , the chance of detecting a real effect.

There is always a trade-off: making  $\alpha$  very small (few false positives) tends to lower power (more missed real effects). Choosing the balance is a design decision, not a mathematical one.

## The workhorse tests

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All follow the same four-step logic; they differ only in the statistic.

- **$t$ -test**, compares means (one sample, or two groups).
- **$\chi^2$  test**, observed vs. expected counts in categories.
- **ANOVA** ( $F$ -test), means across three or more groups.

Each carries assumptions worth stating out loud: independence of observations, and approximate Normality of the relevant averages.

## Worked example 11: significant, or just big?

### Example

Two groups of  $n = 25$  have score means 105 vs. 100, with a pooled SD of 10.

1. Compute Cohen's  $d = \frac{\text{mean}_1 - \text{mean}_2}{\text{SD}}$ .
2. The standard error of the difference is  $\text{SD} \sqrt{1/n_1 + 1/n_2}$ . Compute the  $t$ -statistic. Is the difference significant at 5%?

*Try it yourself first — the solution is on the next slide.*

## Worked example 11: solution

1.  $d = \frac{105 - 100}{10} = 0.5$ , a solidly “medium” effect.
2.  $SE_{\text{diff}} = 10\sqrt{\frac{1}{25} + \frac{1}{25}} = 10\sqrt{0.08} \approx 2.83$ , so  $t = \frac{5}{2.83} \approx 1.77$ , giving  $p \approx 0.08$ : *not* significant at 5%.

### Intuition

A genuine, medium-sized effect can fail the test purely because the sample is too small. Significance depends on both the effect *and*  $n$ , so always report an *effect size* alongside the  $p$ -value.

## The $p$ -value and the replication crisis

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### Link to cognitive science

The  $p$ -value is routinely misunderstood. It is *not* the probability that  $H_0$  is true, nor that the result was “due to chance”. Misuse of  $p < 0.05$ , through multiple comparisons, optional stopping, and publication bias, is a central driver of the **replication crisis** that reshaped psychology and cognitive neuroscience since the 2010s.

Reporting effect sizes and confidence intervals, and pre-registering analyses, are part of the modern remedy.

## From correlation to prediction: regression

Correlation measures the strength of a linear relationship. Regression draws the actual line and lets us *predict*.

### Definition

**Simple linear regression** fits  $\hat{y} = a + bx$  by **least squares**, choosing  $a, b$  to minimise  $\sum(y_i - \hat{y}_i)^2$ . The solution is

$$b = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{\text{Cov}(x, y)}{\text{Var}(x)}, \quad a = \bar{y} - b\bar{x}.$$

# What least squares is doing

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## Intuition

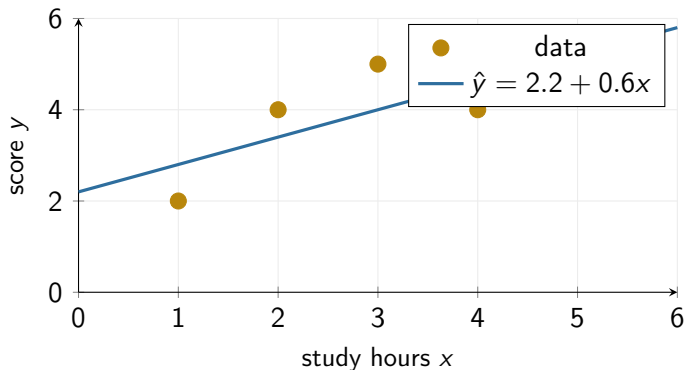
We look for the straight line passing as close as possible to all the points, where “close” is the squared vertical gaps. The slope is just the covariance of  $x$  and  $y$  rescaled by the spread of  $x$ , the same covariance from an hour ago.

## Two facts that make good checks

The residuals always sum to zero and are uncorrelated with  $x$ ; and the line always passes through the mean point  $(\bar{x}, \bar{y})$ .

## The regression line, drawn

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The fitted line through the study-hours data. Each extra hour of study predicts about 0.6 more points.

## Worked example 12: fit the study-hours line

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### Example

For the study-hours data ( $\bar{x} = 3$ ,  $\bar{y} = 4$ ) we already found  $\sum(x - \bar{x})(y - \bar{y}) = 6$  and  $\sum(x - \bar{x})^2 = 10$ , with correlation  $\rho \approx 0.77$ .

1. Find the slope  $b$  and intercept  $a$ .
2. What fraction of the variance in scores does the line explain?

*Try it yourself first — the solution is on the next slide.*

## Worked example 12: solution

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$$b = \frac{6}{10} = 0.6, \quad a = \bar{y} - b\bar{x} = 4 - 0.6(3) = 2.2, \quad \hat{y} = 2.2 + 0.6x.$$

The fraction of variance explained is

$$R^2 = \rho^2 \approx 0.77^2 \approx 0.60,$$

so the line accounts for about 60% of the variation in scores; the remaining 40% is scatter the line cannot capture.

## Worked example 13: predicting firing from contrast

### Link to cognitive science

Fitting firing rate as a function of a stimulus variable is the simplest *neural encoding model*.

### Example

Contrast  $x = (0, 1, 2, 3, 4)$  and firing rate  $y = (1, 1, 2, 2, 4)$ , with  $\bar{x} = \bar{y} = 2$ . Here  $\sum(x - \bar{x})(y - \bar{y}) = 7$  and  $\sum(x - \bar{x})^2 = 10$ .

1. Give the regression line.
2. Predict the firing rate at contrast  $x = 5$ .

*Try it yourself first — the solution is on the next slide.*

## Worked example 13: solution

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$$b = \frac{7}{10} = 0.7, \quad a = 2 - 0.7(2) = 0.6, \quad \hat{y} = 0.6 + 0.7x.$$

At contrast  $x = 5$ :  $\hat{y} = 0.6 + 0.7(5) = 4.1$  spikes/s.

### Common pitfall

This is an *extrapolation* beyond the observed range ( $x \leq 4$ ): trust it only if the linear trend really continues. Real tuning curves eventually saturate, so predicting far outside the data is risky.

# Regression is the atom of modeling

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## Link to cognitive science

Extend regression to many predictors and it becomes *multiple regression*; add a nonlinear “link” and it becomes the *Generalized Linear Model* behind psychometric functions and neural encoding models (Session 5); view it as “fit parameters to minimise error” and it is exactly the objective of supervised machine learning.

And geometrically, least squares is a *projection*, which is where linear algebra picks up the story next session.

### Practice — try it

Four points:  $x = (0, 1, 2, 3)$ ,  $y = (1, 1, 2, 4)$ .

1. Find  $\bar{x}$  and  $\bar{y}$ .
2. Compute  $\sum(x - \bar{x})(y - \bar{y})$  and  $\sum(x - \bar{x})^2$ .
3. Give the regression line  $\hat{y} = a + bx$ .

## Session 2: what we built

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- Summaries: the *mean* (centre), *variance/SD* (spread), and *covariance/correlation* (co-movement).
- The *LLN* and *CLT*: averages converge and become Normal with spread  $\sigma/\sqrt{n}$ , the basis of standard errors and confidence intervals.
- *Hypothesis testing*: *p*-values, the two errors, and why significance is not importance.
- *Regression*: a least-squares line, reusing covariance and pointing ahead to the GLM and machine learning.

Next: linear algebra, vectors, matrices, and geometry.